



Case Report

Toxicological investigations in a fatal and non-fatal accident due to hydrogen sulphide (H₂S) poisoning

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ABSTRACT

Hydrogen sulphide (H₂S) is one of the most toxic natural gas and represents a not rare cause of fatal events in workplaces. We report here a serious accidental poisoning by hydrogen sulphide inhalation involving six sailors. Three of them died while the other three survived and were transported to the emergency room. No greenish discolouration of the body, that could be a feature of these type of deaths, was observed at autopsy. Given that blood and/or urine H₂S detection does not allow to discriminate if it is related to inhalation or to putrefactive processes, the determination of thiosulphate, H₂S main metabolite, is decisive.

The succession of fatal events reported here can be rebuilt by toxicological data interpretation: the subject 1 died after a longer interval of time as demonstrated by the highest blood and urine thiosulfate concentrations; the subject 2 died after a short interval of time as showed by a lower blood and urine thiosulfate concentrations than subject 1; the subject 3 died almost immediately after H₂S inhalation since he showed the lowest blood thiosulfate concentration, and no trace of sulphide and thiosulfate was found in the urine.

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1. Introduction

Hydrogen sulphide (H₂S) is a natural toxic gas that can also induce death. It's an irritant for eyes and airways mucous. In workplaces, like industries of paper, textile, oil refinery, natural gas extraction plants, it is sometimes involved in acute poisoning [1,2]. It is formed by the decomposition of its salts (sulphides) by the action of strong acids or it's naturally produced during the decomposition of organic substances containing sulfurized amino acids. H₂S poisoning is often related to work accidents occurring in confined spaces where the limited entry and exit apertures and the poor natural ventilation cause the decomposition and fermentation of substances producing the toxic gas [3–5]. Outdoor accidental poisoning has been rarely reported [6]. Bassindale et al. [7] and Freireich [8] described some cases of H₂S poisoning happened in geothermal hot pools and in domestic place. H₂S has been detected in suicide cases too [9].

Hydrogen sulphide is recognizable by its characteristic smell of putrefaction; the bond between hydrogen sulphide and haemoglobin,

that lead to sulfohaemoglobin and sulphide methaemoglobin formation, is responsible for the so-called "putrefactive green stain" (chromatic phase of putrefaction) [10].

It is reported that exposure to hydrogen sulphide at 500 ppm rapidly leads to severe intoxication, while at 900 ppm could be immediately fatal [11]. When death does not occur immediately after inhalation, H₂S is metabolized to thiosulfate and it can be detected in blood and urine.

However, in post-mortem evaluations, H₂S detection is not helpful to establish if the cause of death was the exposure to this gas [10,12]. In this case, it is necessary to detect the presence and the concentrations of thiosulfate. In fact, since thiosulfate is produced in the liver by H₂S oxidation, its presence could furnish information about a possible ante-mortem exposure [11,13].

In the present study, the toxicological findings of three fatal and three non-fatal H₂S poisonings occurred to six sailors are reported. The obtained data show the forensic relevance of thiosulfate to discriminate if H₂S blood concentration is related to lethal poisoning or to putrefactive phenomena [12].

2. Cases description

In 2016, during maintenance works on a ship, six sailors were acutely poisoned by inhalation of toxic gases containing hydrogen

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sulphide, due to both organic liquids accumulation and the poor ventilation of the bilge. The bilge was a small recess between the bottom and the left side of the ship into which all water drained, containing liquids that infiltrated the hull and also residues of lubricating oil and fuel. The circumstantial data revealed that the bilge was closed, by a manhole, for many months and no maintenance had been done; the sailors had to pour the liquid from the bilge into the empty space to allow suction by pumps.

Specifically, two sailors (subjects 1 and 2), without any means of protection, opened the door of uncleaned bilge full of putrefactive gas developed from the rotting of waste liquids in there. Before entry in the bilge, these sailors inhaled the gas and, immediately after, lost consciousness. A third sailor (subject 4) tried to help his colleagues, but when he arrived near to bilge door, he lost

consciousness too. Other two sailors (subjects 3 and 5) attempted to rescue their colleagues: the subject 3, coming near the bilge entry, inhaled a large quantity of sulphide; the subject 5 managed to get away with the subject 1 before losing consciousness. The last sailor (subject 6) was informed of the accident and, equipped with an oxygen tank and protective mask, tried to rescue a colleague (subject 4); however, once removed the mask, also the sixth sailor immediately lost consciousness. The circumstantial data revealed that the event occurred about in fifteen minutes. Fig. 1 shows the planimetry of ferryboat where the work accident happened.

Three of sailors (subjects 1–3) died, while the other three were hospitalized and some days later were discharged.

The autopsy of the three died workers were carried out at 24, 26 and 28 h after death, respectively.

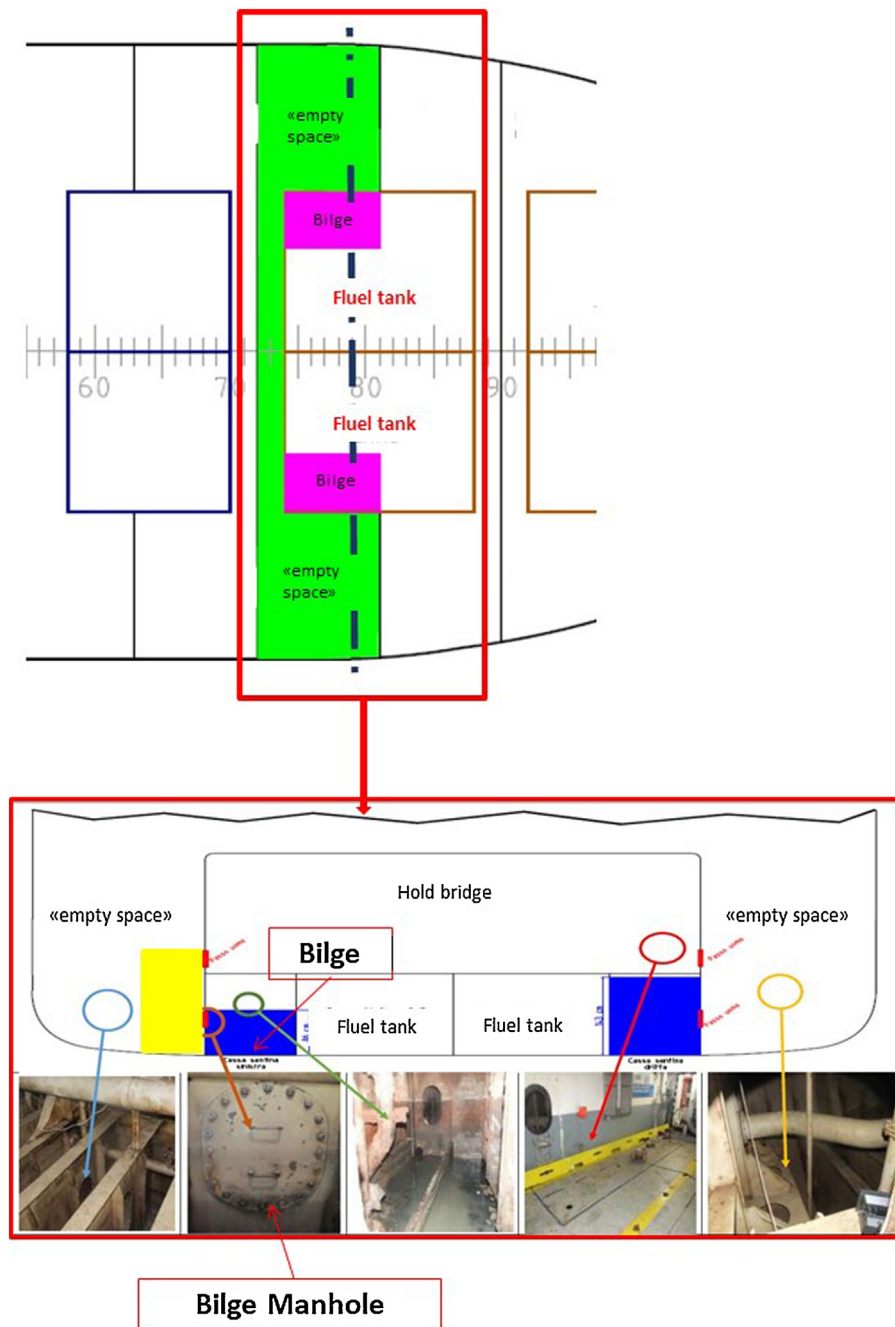


Fig. 1. Planimetry of ferryboat place in which work accident happened. The sailors were rescued in the yellow area in front of the bilge manhole. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

2.1. Subject 1 (male; 29 years-old)

2.1.1. Autopsy findings

Subungual cyanosis; marked hyperemia of conjunctiva; hyperaemia of mucous membrane of mouth, pharynx, esophagus, glottis; red mucous and foamy fluid were found in trachea, and bronchia; lungs were marked swell and congested; marked cerebral edema and congestion were observed.

2.1.2. Histopathologic findings

Marked congestion and slight erosion of tunica propria and submucosa of glottis and trachea; slight erosion and small areas of exudative inflammation of bronchia muscularis mucosae; marked congestion and edema of both lungs; neuronal hypoxic damage, and nuclear pycnosis associated with marked congestion and pericapillary micro-haemorrhages; hypoxic necrosis of epithelium of the convoluted tubules and glomeruli associated with marked congestion of medulla of kidneys.

2.2. Subject 2 (male; 50 years-old)

2.2.1. Autopsy findings

Marked hyperemia of conjunctiva and mouth mucous membrane; hyperemia of glottis mucous membrane; red mucous and foamy fluid in trachea and bronchia; marked swelling and congestion of lungs; marked cerebral edema and congestion.

2.2.2. Histopathologic findings

Marked congestion and small areas of exudative inflammation in tunica propria and submucosa of glottis and trachea; marked congestion and edema of both lungs; marked congestion and slight erosion of esophagus mucous membrane; diffuse neuronal hypoxic damage and nuclear pycnosis associated with marked vessels congestion; hypoxic necrosis of kidneys convoluted tubules epithelium.

2.3. Subject 3 (male; 37 years-old)

2.3.1. Autopsy findings

Marked hyperemia of conjunctiva and mouth mucous membrane; hyperemia of mucous membrane of glottis; red mucous and foamy fluid in trachea and bronchia; marked emphysema and congestion of lungs; marked cerebral edema and congestion.

2.3.2. Histopathologic findings

Marked congestion and edema in both lungs; marked congestion and edema in tunica propria and submucosa of glottis, trachea and bronchia; neuronal hypoxic damage associated with marked congestion of encephalic capillaries and meningeal vessels; hypoxic necrosis of epithelium of the convoluted tubules and glomeruli in kidneys.

The survivor sailors suffered from different type of damages involving the respiratory system and/or heart and/or central nervous system.

2.4. Subject 4 (male; 35 years-old)

At the emergency room (ER) the subject showed severe dyspnoea (oxygen saturation of 81%) and psychomotor agitation; he was intubated and mechanically ventilated; computed axial lungs tomography revealed acute respiratory distress; brain magnetic resonance imaging (MRI) showed two small areas in thalamus due to ischemia. He was hospitalized at the Intensive Care Unit; the subject showed a progressive clinical improvement and after nine days he was discharged. Then, olfactory event-related potentials supported the diagnosis of anosmia.

2.5. Subject 5 (male; 35 years-old)

At the Emergency Room the subject referred headache, sore throat, and dyspnoea; oxygen saturation was of 95%; axial lungs tomography showed widespread inflammation. He was hospitalized at the Medicine Unit. The day after, he was transferred to the Cardiac Intensive Care Unit for chest pain and diffuse T-wave inversions in ECG. Troponins and PCR were increased; the echocardiography showed left ventricular dysfunction (ejection function of 30%). The cardiologists diagnosed toxic myocarditis that, however, showed a progressive improvement; the subject was discharged after 34 days.

2.6. Subject 6 (male; 48 years-old)

At the Emergency Room the subject referred headache, sore throat, dyspnoea; 71 bpm heart rate and 140/70 mmHg blood pressure; moderate leucocytosis; computed axial lungs tomography showed subpleural emphysema. The worker was treated with hydration and antibiotic therapy and discharged after seven days. Spirometry carried out five months later highlighted a restrictive ventilatory defect.

Table 1 reports the main findings observed in the six sailors.

3. Materials and methods

The analyses were carried out on femoral blood and urine samples collected from the corpses of the three deceased sailors, at the autopsies. Femoral blood sample was chosen because it does not have significant post mortem redistribution from central sites [14]. Blood and urine samples were also collected from the three survived sailors the day after hospital admission.

3.1. Instrumentation

The analysis was performed with an HP6890 gas chromatograph (Agilent) interfaced to an HP 5973 Mass Selective Detector (Agilent Technology Inc., Santa Clara, CA) equipped with AT 7683 Series autosampler, operating in the EI mode (70 eV) with SIM monitoring. An HP5 cross-linked fused-silica capillary column (30 m, 0.25 mm i.d.) with a 0.25 μm film thickness (Agilent Technology Inc.) was linked to the mass selective detector (MSD) through a direct capillary interface.

Table 1
Diagnosis of the six sailors after the accident.

Subjects	Dead	Alive	Diagnosis
No. 1	X		Cardiorespiratory arrest
No. 2	X		Heart failure, acute pulmonary oedema, polyvisceral stasis
No. 3	X		Heart failure, acute pulmonary oedema, polyvisceral stasis
No. 4		X	Severe olfactory dysfunction; ischemic thalamic areas
No. 5		X	Toxic myocarditis related to toxic gas inhalation
No. 6		X	Obstructive lung disease related to toxic gas inhalation

The initial temperature of the column oven was held at 100 °C for 2 min and then raised to 220 °C at a rate of 30 °C/min. The temperatures of the injection port and interface were 220 and 210 °C, respectively. Helium was used as a carrier gas at a flow rate of 1.0 mL/min. A splitless injection mode was selected.

3.2. Samples extraction

All reagents and solvents were of analytical grade and were of the highest available purity. Analysis was carried out using the method described by Maebashi et al. [11].

Blood and urine aliquots (0.2 mL) were added to a mixture of 0.5 mL of 20 mM pentafluorobenzyl bromide PFB-Br solution in toluene, 2.0 mL of internal standard solution (10 mM of TBB in ethyl acetate) and 0.8 mL of 5 mM TDMBA solution in oxygen-free water saturated with sodium tetraborate. The preparation was vortexed for 1 min, and 0.1 g of potassium dihydrogen phosphate was added to the mixture. The preparation was again vortexed for 10 s and centrifuged at 2500 rpm for 10 min. An aliquot of the organic phase was then injected into the GC/MS apparatus.

The standard curve for thiosulfate was obtained by plotting the peak area ratio of the derivative (*m/z* 426) of thiosulfate to the TBB (*m/z* 314) versus the thiosulfate concentration. Data acquisition was performed in SIM mode.

4. Results

Sulphide and thiosulphate concentrations ($\mu\text{g/mL}$) found in blood and urine samples of both deceased (subjects 1–3) and survivor sailors (subjects 4–6) are shown in Table 2.

Further toxicological analysis was performed on blood and urine samples of each subject. All specimens resulted negative for alcohol, psychotropic and illegal drugs.

5. Discussion and conclusion

In forensic practice, fatal sulphide intoxications are not unusual in both suicidal and accidental cases, especially in the occupational setting. The autopsy, generally, is not able to furnish clear evidence that H_2S lethal poisoning occurred. The greenish discolouration of the body is often – but not always – present in all H_2S fatal cases [15]. The same consideration concerns the greenish discolouration of the brain or of the viscera that, if present, should only lead to the suspect of H_2S poisoning [16]. Moreover, the gross and microscopic findings are usually unspecific and represented by pulmonary oedema, multiorgan congestion, and erosion of esophagus mucous membrane [17,18].

The toxicological analysis represents the main tool for reliable assessment of sulphide intoxication but, in this field, it is important

to keep in mind that blood H_2S concentration alone could determine an error of evaluation since the impossibility to discriminate if H_2S is due to poisoning or to putrefactive processes [12].

Many researchers have suggested thiosulfate detection as the main investigation to assess sulphide exposure. Thiosulfate is, in fact, a useful and stable marker of hydrogen sulphide exposure [19–21,5].

In the fatal cases reported here, our findings demonstrate H_2S poisoning. Sulphide blood concentration above $0.08 \mu\text{g/mL}$ is enough to lead to death [11]. It is well known that hydrogen sulphide gas inhalation can lead to death because of acute damage to the central nervous system and respiratory depression. Poisoning was not related to room oxygen concentration since the accident happened out of the bilge, in oxygenated place. Even in the presence of oxygen, sulphide poisoning reduces mitochondrial oxygen consumption, inducing asphyxia and leading to death [22].

It is impossible to establish the position of the sailors at the time of the accident because of the rescue operations. However, the succession of fatal events can be rebuilt by toxicological data interpretation.

Subject 1 was the most exposed to sulphide; he died after a longer interval of time as demonstrated by the highest blood and urine thiosulfate concentrations.

Subject 2, inhaled a smaller amount of H_2S and died after a short interval of time as showed by a lower blood and urine thiosulfate concentrations than subject 1.

The subject 3 died almost immediately after H_2S inhalation; in fact, he showed the lowest blood thiosulfate concentration, and no trace of sulphide and thiosulfate was found in the urine.

Toxicological data appeared to be most informative compared to macroscopic and histological findings. In fact, the gross and microscopic examination showed similar findings in the three subjects revealing multi-organ cytotoxic damage, observed, especially, in brain, lungs and kidneys. Since that H_2S environmental levels were very high, in the cases presented here the events happened in a very short interval of time; no histological differences related to toxic effects were revealed; whereas toxicological findings seem to be the best way to reconstruct the accident.

Post-mortem blood and urine thiosulfate determination is helpful in forensic investigation concerning H_2S poisoning. Thiosulfate detection is of primary importance to discriminate if H_2S blood concentration is related to lethal poisoning or to putrefactive phenomena [12]. Moreover, blood and urine thiosulfate concentrations can indicate the chronological order of each death.

The toxicological analysis on survivor workers resulted negative, except in a sailor (subject 4) in which a low blood and urine thiosulfate concentration was detected. This evidence suggests that H_2S metabolism is very fast. Among survivors, the worker resulted positive to thiosulfate had the most serious H_2S -related damages characterized by acute respiratory distress, low oxygen saturation (81%), two ischemic thalamic areas and anosmia, suggesting the exposure to a greater level of H_2S compared to other survivor sailors.

The presented case highlights that the analysis of H_2S and, in particular, of thiosulfate can be very useful in forensic evaluation of H_2S poisoning providing information on survival time and event dynamic.

Finally, it must be highlighted that several cases of work-related H_2S intoxication have been reported in Literature. This demonstrates that it is necessary to implement greater risks awareness from workers, specific training, safe and shared work procedures, availability of suitable equipment. Every initiative aimed to evaluate the improvement of the workplace in which accident occurred and to increase the awareness about works with a high risk of H_2S poisoning (“learning from error”) could be very useful.

Table 2

Blood and urine sulphide and thiosulphate concentrations ($\mu\text{g/mL}$) found in dead and survivor subjects.

	Sample	Sulphide ($\mu\text{g/mL}$)	Thiosulphate ($\mu\text{g/mL}$)
Subject 1	Blood	9.3	45.7
	Urine	14.7	510.1
Subject 2	Blood	5.3	35.5
	Urine	9.7	10.4
Subject 3	Blood	5.9	21.5
	Urine	n.d.	n.d.
Subject 4	Blood	n.d.	6.9
	Urine	n.d.	10.8
Subject 5	Blood	n.d.	n.d.
	Urine	n.d.	n.d.
Subject 6	Blood	n.d.	n.d.
	Urine	n.d.	n.d.

n.d. Not detected.

CRediT authorship contribution statement

Elvira Ventura Spagnolo: Investigation. **Guido Romano:** Methodology. **Pietro Zuccarello:** Writing - review & editing. **Antonino Laudani:** Formal analysis. **Cristina Mondello:** Writing - original draft. **Antonina Argo:** Conceptualization. **Stefania Zerbo:** Visualization. **Nunziata Barbera:** Supervision.

References

- [1] A. Christia-Lotter, C. Bartoli, M.D. Piercecchi-Marti, D. Demory, A.L. Pelissier-Alicot, A. Sanvoisin, G. Leonetti, Fatal occupational inhalation of hydrogen sulphide, *Forensic Sci. Int.* 169 (2–3) (2007) 206–209.
- [2] J.A. Nantel, Un poison violent, l'hydrogène sulfure, 13, *Bulletin d'Information Toxicologique*, 1997.
- [3] C. Zaba, J.T. Marcinkowski, A. Wojtyła, A. Tężyk, J. Tobolski, Z. Zaba, Acute collective gas poisoning at work in a manure storage tank, *Ann. Agric. Environ. Med.* 18 (2) (2011) 448–451.
- [4] H. Kfir, S. Rimbrot, A. Markel, Toxic effects of hydrogen sulphide: experience with three simultaneous patients, *QJM* 108 (12) (2015) 977–978.
- [5] D. Poli, B. Solarino, G. Di Vella, L. Tattoli, G. Strisciullo, M. Goldoni, A. Mutti, R. Gagliano-Candela, Occupational asphyxiation by unknown compound(s): environmental and toxicological approach, *Forensic Sci. Int.* 197 (1–3) (2010) e19–e26.
- [6] S. Nogué, R. Pou, J. Fernández, P. Sanz-Gallén, Fatal hydrogen sulphide poisoning in unconfined spaces, *J. Occup. Med. Toxicol.* 61 (3) (2011) 212–214.
- [7] T. Bassindale, M. Hosking, Deaths in Rotorua's geothermal hot pools: hydrogen sulphide poisoning, *Forensic Sci. Int.* 207 (1–3) (2011) e28–9.
- [8] A.W. Freireich, Hydrogen sulphide poisoning: report of two cases, one with fatal outcome, from associated mechanical asphyxia, *Am. J. Pathol.* 22(1)(1946) 147–155.
- [9] R.N. Sams, H.W. Carver 2nd, C. Catanese, T. Gilson, Suicide with hydrogen sulphide, *Am. J. Forensic Med. Pathol.* 34 (2) (2013) 81–82.
- [10] N. Barbera, A. Montana, F. Indorato, N. Arbouche, G. Romano, Domino effect: an unusual case of six fatal hydrogen sulphide poisonings in quick succession, *Forensic Sci. Int.* 260 (2016) e7–e10.
- [11] K. Maebashi, K. Iwadate, K. Sakai, A. Takatsu, K. Fukui, M. Aoyagi, E. Ochiai, T. Nagai, Toxicological analysis of 17 autopsy cases of hydrogen sulphide poisoning resulting from the inhalation of intentionally generated hydrogen sulphide gas, *Forensic Sci. Int.* 207 (1–3) (2011) 91–95.
- [12] N. Barbera, A. Montana, F. Indorato, N. Arbouche, G. Romano, Evaluation of the role of toxicological data in discriminating between H₂S femoral blood concentration secondary to lethal poisoning and endogenous H₂S putrefactive production, *J. Forensic Sci.* 62 (2) (2017) 392–394.
- [13] R.O. Beauchamp Jr, J.S. Bus, J.A. Popp, C.J. Boreiko, D.A. Andjelkovich, A critical review of the literature on hydrogen sulphide toxicity, *Crit. Rev. Toxicol.* 13 (1) (1984) 25–97.
- [14] V.M. Hargrove, J.R. McCutcheon, Comparison of drug concentrations taken from clamped and unclamped femoral vessels, *J. Anal. Toxicol.* 32 (8) (2008) 621–625.
- [15] A.K. Chaturvedi, D.R. Smith, D.V. Canfield, A fatality caused by accidental production of hydrogen sulphide, *Forensic Sci. Int.* 123 (2–3) (2001) 211–214.
- [16] S.H. Park, Y. Zhang, J.J. Hwang, Discolouration of the brain as the only remarkable autopsy finding in hydrogen sulphide poisoning, *Forensic Sci. Int.* 187 (1–3) (2009) e19–e21.
- [17] S. Kage, S. Ito, T. Kishida, K. Kudo, N. Ikeda, A fatal case of hydrogen sulphide poisoning in a geothermal power plant, *J. Forensic Sci.* 43 (4) (1998) 908–910.
- [18] C. Sastre, V. Baillif-Couniou, P. Kintz, V. Cirimele, C. Bartoli, M.A. Christia-Lotter, M.D. Piercecchi-Marti, G. Leonetti, A.L. Pelissier-Alicot, Fatal accidental hydrogen sulphide poisoning: a domestic case, *J. Forensic Sci.* 58 (1) (2013) S280–S284.
- [19] S. Kage, H. Ikeda, N. Ikeda, A. Tsujita, K. Kudo, Fatal hydrogen sulphide poisoning at a dye works, *Leg. Med. (Tokyo)* 6 (3) (2004) 182–186.
- [20] S. Kage, K. Takekawa, K. Kurosaki, T. Imamura, K. Kudo, The usefulness of thiosulfate as an indicator of hydrogen sulphide poisoning: three cases, *Int. J. Legal Med.* 110 (4) (1997) 220–222.
- [21] J. Kangas, H. Savolainen, Urinary thiosulphate as an indicator of exposure to hydrogen sulphide vapour, *Clin. Chim. Acta* 164 (1) (1987) 7–10.
- [22] J. Jiang, A. Chan, S. Ali, A. Saha, K.J. Haushalter, W.L. Lam, M. Glasheen, J. Parker, M. Brenner, S.B. Mahon, H.H. Patel, R. Ambasudhan, S.A. Lipton, R.B. Pilz, G.R. Boss, Hydrogen sulphide–Mechanisms of toxicity and development of an antidote, *Sci. Rep.* 6 (2016) 20831.